

ML Assignment

1. Implement Expectation–Maximization (EM) Algorithm for Classification

Consider the following 1-D dataset: $X = \{1, 2, 8, 9\}$. Assume the data is generated from a mixture of two Gaussian distributions ($K = 2$). Initial parameters are Mixing coefficients: $\pi_1 = 0.5$, $\pi_2 = 0.5$, Means: $\mu_1 = 2$, $\mu_2 = 8$, Variances: $\sigma_1^2 = 1$, $\sigma_2^2 = 1$

2. EM Algorithm for Gaussian Mixture Model (Binary Classification)

Consider the following 1-D dataset: $X = \{2, 3, 4, 10, 11, 12\}$. Assume the data is generated from a mixture of two Gaussian distributions ($K = 2$). Initial parameters: such as Mixing coefficients: $\pi_1 = 0.5$, $\pi_2 = 0.5$, Means: $\mu_1 = 3$, $\mu_2 = 11$ Variances: $\sigma_1^2 = 1$, $\sigma_2^2 = 1$